



Péter Juhász

PhD Researcher in Mathematics

CONTACT

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- shepherd92

EDUCATION

2022 – 2025: PhD in Mathematics
AU, Denmark

2019 – 2021: MBA in Finance
BUTE, Hungary

2013 – 2016: MSc in Physics
BUTE, Hungary

2013 – 2016: BSc in Physics
BUTE, Hungary

CERTIFICATIONS

2021: Software Architect
ISAQB

2021: Software Designer
Bosch Group

SKILLS

Stochastic Processes

Probability

Python

C++

SQL

GIT

LANGUAGES

Hungarian

English

German

PROFILE

I am a PhD Researcher in Mathematics at Aarhus University specialized in stochastic modeling and complex network analysis, finishing in December 2025. My research involves the application of probabilistic methods to complex systems and developing efficient C++/Python algorithms. Previously, I worked at Bosch as a deep learning researcher and software architect in automated driving, focusing on perception, prediction, decision-making, and scalable software design.

PROFESSIONAL EXPERIENCE

PhD Researcher

Aarhus University, Denmark

2022 – 2025

- Developed and analyzed stochastic models for complex, high-dimensional systems.
- Designed efficient C++ and Python algorithms for stochastic network models.
- Proved theoretical results in stochastic point processes and random graphs.

Deep Learning Researcher

Bosch Group, Hungary

2021 – 2022

- Patented a stochastic trajectory prediction model improving accuracy by 3× and uncertainty estimation by 10× over baseline.
- Developed object detection models using U-Net architecture and ultrasonic sensor data.

Software Architect

Bosch Group, Hungary

2018 – 2021

- Led a team of 11 developing environment prediction modules for automated driving as a technical lead.
- Created a framework for automated ground-truth generation.
- Oversaw 2 teams in the integration of driver monitoring systems.

Software Engineer

Ericsson Telecommunications, Hungary

2016 – 2018

- Designed and optimized a response time-based load-balancing algorithm for distributed database servers.
- Coordinated a team of 8 as a deputy of the product owner.
- Improved response times with custom load-balancing heuristics.

AWARDS, GRANTS

2025: Individual Travel Grant
European Mathematical Society
500 EUR

2024: Academic Mobility Grant
Aarhus University
20 000 DKK

2022: PhD Fellowship
Danish Data Science Academy
1 890 000 DKK

2021: Excellence Award
Hungarian National Bank
350 000 HUF



SELECTED PUBLICATIONS, PATENTS

- C. Hirsch, B. Jahnel, and P. Juhász.
Functional limit theorems for edge counts in dynamic random connection hypergraphs,
arXiv preprint (2025), arxiv:2507.16270
- P. Juhász.
Method and device for predicting the trajectory of a traffic participant, and sensor system,
CNIPA: CN113988353A, EPO: EP3916697A1
JPO: JP2021190119A USPTO: US2021366274A1 (2021)
- P. Juhász.
Information propagation in stochastic networks,
Phys. A (577) (2021), 126070
- P. Juhász, J. Stéger, D. Kondor, and G. Vattay.
A Bayesian approach to identify Bitcoin users,
PLoS One (13) (2018), e0207000

TEACHING

Advanced Probability Theory
10 ECTS Exercise Class

2025

Data Project
10 ECTS Computer Project Class

2024

Ordinary Differential Equations
5 ECTS Exercise Class

2023